Scientific Research in Information Systems: A Beginner's Guide (2nd edition)Teaching Materials

Created by Professor Jan Recker

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Course Overview

- This course will provide an introduction and overview of scientific research in Information Systems. It will discuss elementary scientific principles of research, explore key decisions in identifying research questions, designing a study, developing and using theory, research methods, and discuss challenges in paper writing, publishing, and scientific ethics.

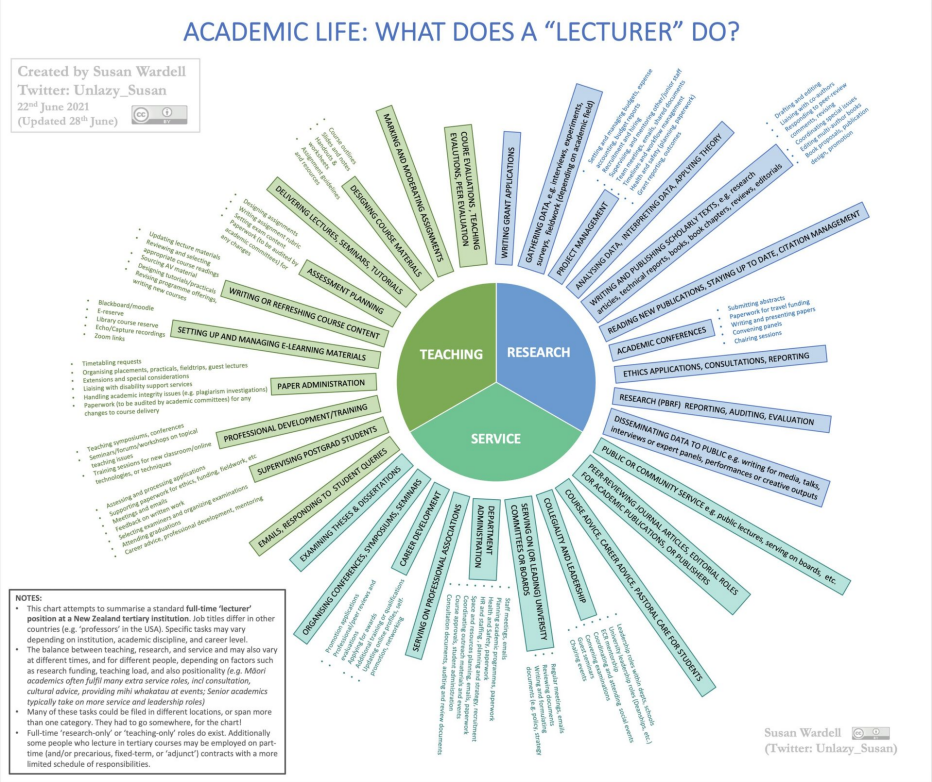
- The course will be taught in an interactive style where we jointly discuss learnings, critique articles, attempt to apply what we have learned to improve and extend our own research and publications, and work together to reach a deeper understanding of how to produce significant Information Systems research. The course will feature readings, lectures as well as student presentations and discussions in the classroom.

Learning Outcomes

- **Students learn to…**
  - ...understand principles of scientific inquiry such as replicability, independence, and precision.
  - …evaluate phenomena and research problems of interest to information systems scholars
  - ...apply discipline and technical knowledge to analyze and evaluate scientific processes and outcomes in information systems.
  - … distinguish, evaluate, and compare different research methods used in information systems research
  - …frame important research questions and design research studies appropriate to address these research questions.
  - …develop written communication skills to structure, explain and defend scientific thinking.
  - …appreciate ethical issues in planning, conduct, and publishing of information systems research.

Chapter 1:
Introduction to Information Systems Research

What does a professor do?



Let‘s get going:
What is Research?

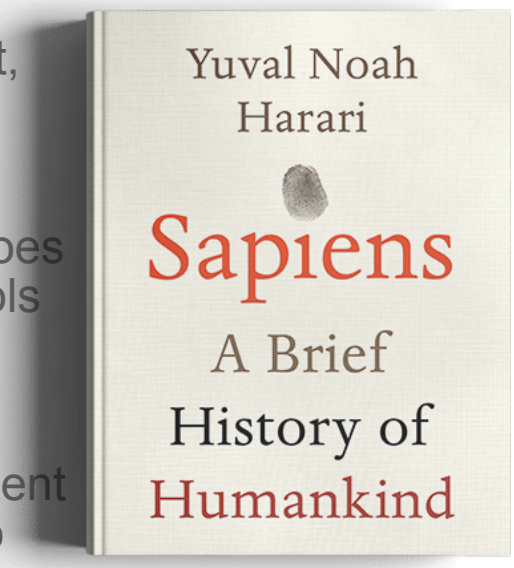
What is a famous study you have heard of?

The Scientific Revolution: Mankind realizes they do not know everything.

a) **The willingness to admit ignorance**. Science assumes that we don't know everything. Even more critically, it accepts that the things that we think we know could be proven wrong as we gain more knowledge. No concept, idea or theory is sacred and beyond challenge.

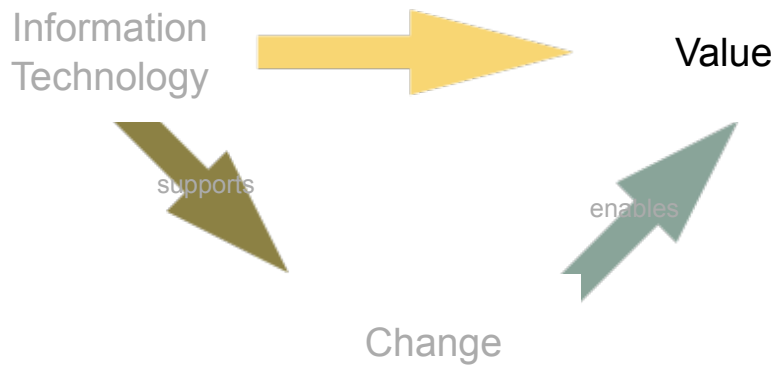
b) **The centrality of observation**. Having admitted ignorance, science aims to obtain new knowledge. It does so by gathering observations and then using formal tools such as logic or mathematics to connect these observations into comprehensive theories.

c) **The acquisition of new powers**. Science is not content with creating theories. It uses these theories in order to acquire new powers, and in particular to develop new technologies.



What are we researching in IS?

- What is Information Systems research about?



Definition of IS research

- The information systems research discipline is concerned with examining the **development, use, and impact** of digital information and communication technology.

Two key aspects of that definition (1)

- A focus on digital technologies, that is, technologies that rely on **digitisation**, the encoding of analogue information into digital format (essentially, bits and bytes that contain 0s and 1s).

- This makes IS research dynamic and fast-moving: these artefacts change all the time!

Two key aspects of that definition (2)

- The study of **development, use, and impacts** is very broad and heterogenous:
  - What is development
  - What do we mean by use?
  - What could be impacts?

- This makes IS research diverse and challenging: different theories and methods are needed to study all these aspects!

There are many ways of doing research on Information Systems

- **Scholars in information technology, software engineering, and computer science** study the technical and computational attributes of digital technology as such.

- **Scholars in behavioural, cognitive, and psychological sciences** study individuals' exposure, use, appropriation, and general behaviours [within digital technology domains].

- **Scholars in organisational science, management, and business** study how corporate environments shape, and are shaped by, digital technology.

- **Economists** study the large-scale effects of digital technology diffusion and innovation on organisations, markets, and societies.

Types of research

- **Conceptual research**
  - is concerned with an analysis of a problem at hand without a commitment to any (technologically-driven) solutions

- **Formal research**
  - Is concerned with describing concepts in mathematical terms (so as to remove any potential ambiguity with respect to their interpretation) and presenting theorems with corresponding proofs to provide more insight into these concepts

Types of research

- **Technological research**
  - Is concerned with the development of sophisticated software artifacts to prove that certain ideas can indeed be realized and to reason about the comparative strengths and weaknesses of various architectural approaches
  - E.g., Workflow Systems (YAWL) or Process Mining (Celonis)

- **Experimental research**
  - typically addresses a limited set of properties of a phenomenon (e.g., a property of a technology or person, or a method), and examine these properties in controlled settings
  - E.g., experiments in medicine or physics

Types of research

- **Empirical research**
  - sets off to study phenomena in real-life practice, and to build or confirm theories about how the phenomena came to be about.
  - can be based on case studies, surveys, action research, grounded theory, or other methods.
  - E.g., Technology Acceptance, Success factor models

What do you need to do when you pursue research in your studies?

- Demonstrate your ability to conduct research and scholarship that make a unique contribution and meet the standards of **credibility** and **verifiability**

- Example: PhD

  “The Doctor of Philosophy degree is awarded in recognition of a student's erudition in a broad field of learning and for notable accomplishment in that field through an **original and substantial contribution to [the body of] knowledge**.”

From the QUT Doctor of Philosophy Regulations (IF49)

How do you know you are not on the right track?

- Common misleading views on what research is about:
  - **The great idea**

    “I have just had this great idea! I do not know if anyone else has ever had the same idea, because I’ve not checked, and I’m rather new in this field. Anyway, my idea is brilliant, so I really would like to share it with you all.”

How do you know you are not on the right track?

- Common misleading views on what research is about:
  - **Other people's idea**

    "I have just read this great book that I really like a lot. I'll just give you a short resume of the interesting points in the book and apply it to this situation over here."

How do you know you are not on the right track?

- Common misleading views on what research is about:
  - **The software hacker**

    "I have just built this great computer system/software tool/mobile application. It is not based on previous theories or empirical findings. I am not very theoretical myself, but the system has a lot of fantastic features, and the interface is neat. Plus, people could really use it."

How do you know you are not on the right track?

- Common misleading views on what research is about:
  - **The theory hacker**

    "I have come up with this theory/conceptual framework/model/methodology. It is not related to other theories/conceptual frameworks/models, or any empirical data for that matter. Most of the concepts have been defined differently by all the big shots in the field, but I just do not like their categories so I have invented my own. And I think it is surely better (I haven't checked that either)."

What matters to research?

- **Ambition**
  - The ambition to complete a doctoral degree to enter (or progress) an academic career is different from the ambition to pursue a research degree as recognition of thought leadership for industry or corporate careers.
  - **What is your ambition?**

- **Dedication**
  - The enthusiasm that you need to work on a novel and intrinsically complex challenge for a substantial amount of time
  - **Are you dedicated?**

- **Commitment**
  - The willingness to free time and resources to work dedicatedly on the research
  - **Are you free to commit yourself?**

Research is a challenging task

- Fewer than 65 percent of people who start PhD programs finish them (Bowen, W. and N. Rudenstine. 1992). More recent data paints a similar picture: Attrition in residential doctoral programs is as high as 50 percent in face-to-face programs (de Vise, 2010) and 50–70 percent in online programs (Rigler Jr. et al., 2017). In Germany, attrition rates are said to be about 15 percent of doctoral students in social science disciplines and about 25 percent in engineering disciplines (Franz, 2015).

- Ten percent of doctoral students claim to have considered suicide, and 54 percent of doctoral students have felt so depressed at various stages that they had difficulty functioning (OnlinePhDPrograms.Net).

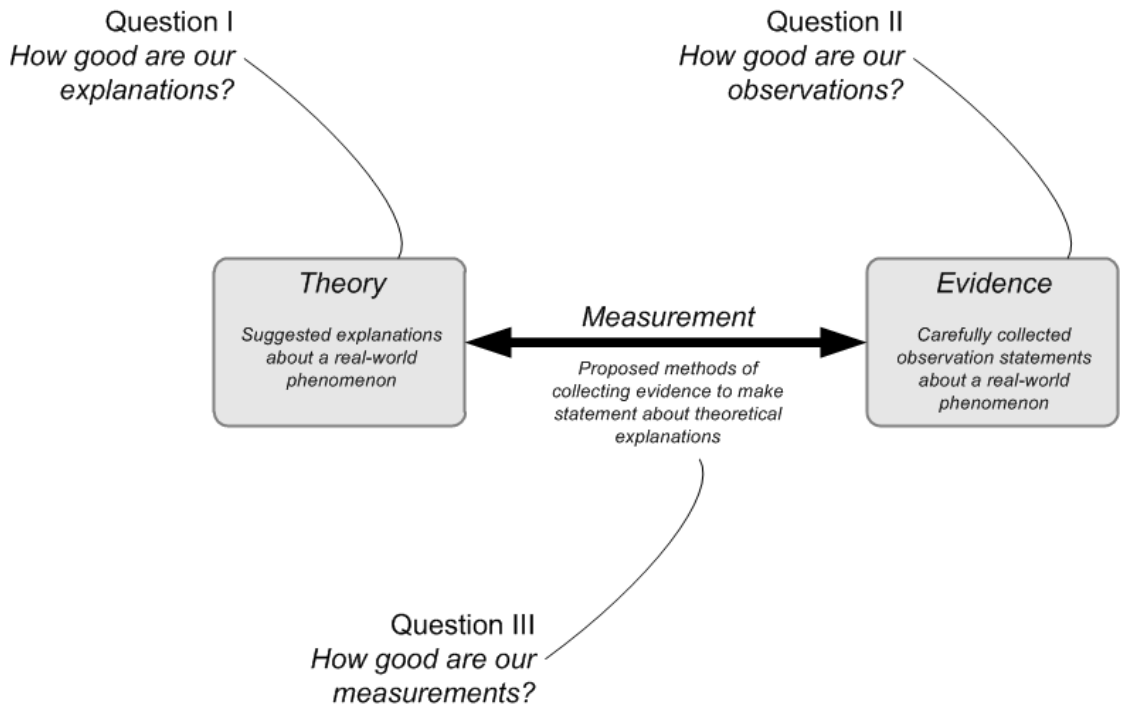
- In one study, 43 percent of participating graduate students reported experiencing more stress than they could handle, with PhD students expressing the greatest stress. More than half listed stress or burnout as a major concern, about a quarter cited feeling like outsiders, and nearly a third listed their relationships withprofessors as a cause of stress (Patterson, 2016).

- Why is that?

Instead of taking in more knowledge, you have to create knowledge

- Most common reason is that students do not understand the formal cause of earning a research degree.
  - Elementary school is like learning to ride a tricycle.
  - High school is like learning to ride a bicycle.
  - College is like learning to drive a car.
  - A master's degree is like learning to drive a race car.
  - Students often think that the next step (a research degree) is more of the same, like learning to fly an airplane.
  - On the contrary, a research degree (like a PhD) is like learning to design a new car.

How do you create new knowledge?

- The body of knowledge is the current accumulation of **theories**, **evidence** and **methods** in a certain domain (e.g. medicine, management, education etc).
- Typically consists of theories that have been evidenced – i.e., not falsified (yet).
- Also consists of methods that have been used to evidence or falsify theories.
- Sometimes can be innovative and important new evidence.
- Is available in the scientific community in the form of paper, articles and books.

What is the body of knowledge?